Course Code: B23BS2101

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R23**

II B.Tech. I Semester MODEL QUESTION PAPER

DISCRETE MATHEMATICS AND GRAPH THEORY

(Common to CSE, AIML, CSBS, IT, AIDS, CSG, CIC, CSIT)

Time: 3 Hrs. | **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	**KL**	**M**
1.	**a).**	Write the inverse, converse and contra positive of " If $\triangle ABC$ is a right angle triangle then $AC^2 = AB^2 + BC^2$ "	**1**	**2**	**2**
	b).	Translate the following statement into symbolic form. "Any integer is either positive or negative".	**1**	**2**	**2**
	c).	Define relation matrix and give an example.	**2**	**1**	**2**
	d).	Define Lattice and give an example.	**2**	**1**	**2**
	e).	How many 4-digit numbers divisible by 5 can be formed using the digits 3, 7, 1, 5, 6.	**3**	**2**	**2**
	f).	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 0, n \geq 2$.	**3**	**3**	**2**
	g).	Define adjacency matrix.	**4**	**1**	**2**
	h).	Define Eulerian graph.	**4**	**1**	**2**
	i).	Define tree and give an example.	**5**	**1**	**2**
	j).	Define Graph coloring.	**5**	**1**	**2**

5 x 10 =50Marks

		UNIT-1			
2.	**a).**	Prove that $\{[(p \vee q) \rightarrow r] \wedge \neg p\} \rightarrow (q \rightarrow r)$ is a tautology.	**1**	**3**	**5**
	b).	Verify that the following argument is valid by using the rules of inference If Clifton does not live in France, then he does not speak French. Clifton does not drive a Datsun. If Clifton lives in France, then he rides a bicycle. Either Clifton speaks French, or he drives a Datsun. Hence, Clifton rides a bicycle.	**1**	**3**	**5**
		OR			
3.	**a).**	Verify that the following argument is valid by using the rules of inference, quantifiers. Babies are illogical.	**1**	**3**	**5**

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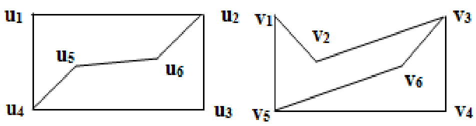
		Nobody is despised who can manage a crocodile. Illogical people are despised. Hence, babies cannot manage crocodiles.			
	b).	Find the PDNF and PCNF of $p \vee \neg q$	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Find the number of integers between 1 and 250 which are divisible by any of the integers 2, 3, 5 or 7.	**2**	**3**	**5**
	b).	Let R denote a relation on the set of ordered pairs of positive integers such that $(x, y)R(u, v)$ *if and only if* $xv = yu$. Then establish that 'R' is an equivalence relation.	**2**	**3**	**5**
		OR			
5.	**a).**	Define Hasse diagram and draw Hasse diagram of $\left(P(A), \subseteq\right)$, where A = {1, 2, 3}.	**2**	**3**	**5**
	b).	Define bijective and inverse function. Find the inverse of the function $f(x) = 5x + 2$.	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	A cricket team of 11 is to be selected out of 14 players of whom 5 are bowlers. Find the number of ways in which this can be done so as to include atleast 3 bowlers.	**3**	**3**	**5**
	b).	i) Determine the term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$ ii) Determine the coefficient of $x^5y^{10}z^5w^5$ in the expansion $(x + 7y + 3z + w)^{25}$	**3**	**3**	**5**
		OR			
7.	**a).**	How many integral solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ where $x_1 \geq 3, x_2 \geq 2, x_3 \geq 4, x_4 \geq 6, x_5 \geq 0$.	**3**	**3**	**5**
	b).	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 0,\ n \geq 2$ using generating functions.	**3**	**3**	**5**
		UNIT-4			
8.		Define isomorphism of graphs and determine the Isomorphism between the following graphs.  	**4**	**3**	**10**
		OR			
9.	**a).**	Prove that in any graph, there is an even number of vertices of odd degree.	**4**	**3**	**5**

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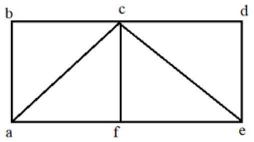
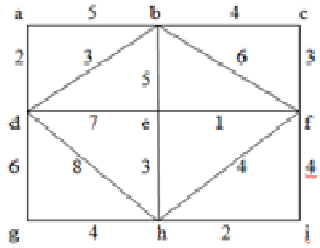
	b).	Define Hamiltonian graph and find the Hamiltonian path and Hamiltonian graph in the following graph. 	4	3	5
		UNIT-5			
10.	**a).**	State and Prove Euler's formula for planar graphs.	**5**	**3**	**5**
	b).	Show that a tree with n vertices has exactly (n-1) vertices.	**5**	**3**	**5**
		OR			
11.		Find the minimal spanning tree for the following weighted graph 	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23HS2101

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R23**

II B.Tech. I Semester MODEL QUESTION PAPER

UNIVERSAL HUMAN VALUES-2: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT

(Common to all programmes of Engineering)

Time: 3 Hrs. **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	**a).**	What are the basic guidelines for value education?	1	2	2
	b).	What is MBTI personality test?	1	2	2
	c).	How can we differentiate between the needs of the Self and the needs of the Body?	2	2	2
	d).	What are the characteristics and activities of the Self (I)?	2	2	2
	e).	How is 'respect' defined in the context of human interaction?	3	2	2
	f).	How is society described in relation to the family?	3	2	2
	g).	How are the four orders of nature interconnected?	4	2	2
	h).	How does the idea of self-regulation in nature contribute to its harmony?	4	2	2
	i).	Define definitiveness of (ethical) human conduct.	5	2	2
	j).	Explain how humanistic education can influence professional ethics.	5	2	2

5 x 10 = 50 Marks

		UNIT - I	CO	KL	M
2.	**a).**	Discuss natural acceptance	1	2	5
	b).	Differentiate prosperity and deprivation	1	2	5
		OR			
3.	**a).**	Deliberate the right understanding in perspective to self exploration.	1	2	5
	b).	What are the key functions of the MBTI? Explain.	1	2	5
		UNIT - II			
4.	**a).**	Illustrate coexistence of "I" and "Body".	1	2	5
	b).	Distinguishing between the Needs of the Self and the Body	1	2	5
		OR			
5.	**a).**	Discuss Characteristic activities of Harmony with "I".	1	2	5
	b).	Explain Sanyam and Health.	1	2	5

		UNIT - III			
6.	**a).**	Write a note on human-human relationship as regarding harmony.	**2**	**2**	**5**
	b).	Differentiate intention and competence.	**2**	**2**	**5**
		OR			
7.	**a).**	Discuss salient values in relationship.	**3**	**2**	**5**
	b).	Illustrate universal Harmonious Society - an Undivided society.	**3**	**2**	**5**
		UNIT - IV			
8.		Discuss orders of life in nature and its significance self regulation of individual	**4**	**2**	**10**
		OR			
9.		Illustrate existence of human being as coexistence with universe in perspective of space	**4**	**2**	**10**
		UNIT - V			
10.		Discuss importance of professional competence for augmenting universal human order.	**5**	**2**	**10**
		OR			
11.	**a).**	Case study of typical holistic technologies.	**5**	**2**	**5**
	b).	Role of engineer in promoting harmony in society	**5**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

SRKR ENGINEERING COLLEGE

Course Code: B23IT2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
II B.Tech. I Semester MODEL QUESTION PAPER					
DATABASE MANAGEMENT SYSTEMS					
Common to IT, AIDS & CSBS					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	**KL**	**M**
1.	**a).**	List any four advantages of Database Systems.	**1**	**1**	**2**
	b).	Describe unary and binary relationships in E-R diagram.	**2**	**2**	**2**
	c).	Explain the use of primary key with an example.	**2**	**1**	**2**
	d).	Describe select (σ) operator in relational algebra with an example.	**3**	**2**	**2**
	e).	Explain the use of "*in*" operator in SQL.	**3**	**1**	**2**
	f).	In what way, a view and a base table are different.	**2**	**2**	**2**
	g).	Explain the purpose of Normalization.	**4**	**2**	**2**
	h).	How is dependency preservation checked after decomposition?	**4**	**1**	**2**
	i).	Explain atomicity of a transaction with an example.	**5**	**1**	**2**
	j).	Draw state diagram of a Transaction	**5**	**1**	**2**
5 x 10 = 50 Marks					
		UNIT-1			
2.	**a).**	Compare database system with file system.	**1**	**2**	**5**
	b).	Describe various mapping cardinalities in E-R diagrams with suitable examples.	**2**	**2**	**5**
		OR			
3.	**a).**	Explain the function of each component in DBMS architecture.	**1**	**2**	**5**
	b).	Model an E-R diagram for Library Information System by considering the activities in the system.	**2**	**3**	**5**
		UNIT-2			
4.	**a).**	Explain DDL and DML Commands with examples.	**2**	**2**	**5**
	b).	Make use of E-R model concepts to convert the following E-R diagram into a collection of relations.	**2**	**3**	**5**

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		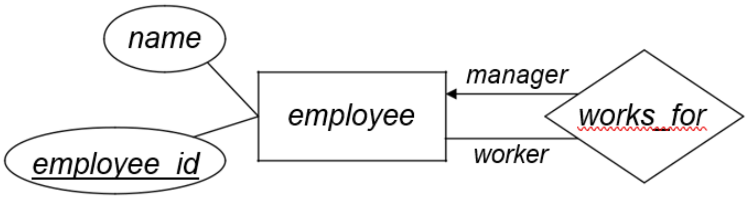			
		OR			
5.	**a).**	Describe foreign key constraint and unique constraint with suitable examples.	**2**	**2**	**5**
	b).	Consider the following relational database: Emp (**<u>ename</u>**, street, city) , Works (**<u>ename</u>**, company-name, salary), company (**<u>company-name</u>**, city), manages(**<u>ename</u>**, manager-name) Use relational algebra operators to answer the following. i) Find the names and cities of all employees who work for First Bank Corporation ii) Find names, street, and cities of all employees who work for First Bank Corporation and earn more than 10,000 iii) Find all employees in the database who earn more than every employee of Small Bank Corporation	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	What is the difference between a nested query and a correlated query?	**3**	**2**	**4**
	b).	Consider the following database: Students (**S_ID**, Dept, CGPA) Courses (**C_ID**, Offered_By_Dept, Credits) Enrolled (**S_ID, C_ID**, Grade) Apply SQL concepts to answer the following queries. i) Find the best CGPA in each department. ii) Find the IDs of students who enrolled for some course offered by other department. Display student dept and course offering department along with S_ID. iii) Find the IDs of students who got at least two “A” grades.	**3**	**3**	**6**
		OR			
7.	**a).**	Explain various types of joins in SQL	**3**	**2**	**5**
	b).	Explain the use of *group by* and *having* clauses with an example.	**3**	**2**	**5**
		UNIT-4			
8.	**a).**	Define lossless join decomposition. Suppose that we decompose the schema R(A,B,C,D,E) into R1(A,B,C) and R2(A,D,E). **Determine** whether this decomposition is a lossless decomposition under the following functional dependencies: F={A→BC,BD→E,B→D,E→A}	**4**	**3**	**5**
	b).	Determine that a relation in BCNF is also in 3NF and not vice-versa with a suitable example.	**4**	**3**	**5**
		OR			

9.	**a).**	Explain Fourth Normal form with an example	**4**	**2**	**5**
	b).	Let R (ABCDE) F={A→B, BC→D, D→E). Find all candidate keys of R and also determine the highest normal of R.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Explain ARIES recovery algorithm.	**5**	**2**	**5**
	b).	Explain Primary Indexing	**6**	**2**	**5**
		OR			
11.	**a).**	Explain 2-Phase Locking Protocol.	**5**	**2**	**5**
	b).	Demonstrate the construction of a B+ tree of order 3 for the following key values : 20,15,10,5,8,30,1,40,35,25	**6**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980 SRKR ENGINEERING COLLEGE AUTONOMOUS

					Course Code: B23IT2102
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
II B.Tech. I Semester MODEL QUESTION PAPER					
OBJECT-ORIENTED PROGRAMMING THROUGH JAVA					
Common to IT, AIDS & CSBS					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 x 2 = 20 Marks

			CO	KL	M
1.	**a).**	What is the purpose of the main method in a Java program?	**1**	**1**	**2**
	b).	Explain the difference between ++i and i++ in Java.	**1**	**2**	**2**
	c).	What is the significance of the this keyword in Java?	**2**	**2**	**2**
	d).	Explain method overloading with an example.	**2**	**3**	**2**
	e).	How do you declare a two-dimensional array in Java? Provide a code example.	**3**	**3**	**2**
	f).	What is the purpose of the super keyword in Java inheritance?	**3**	**2**	**2**
	g).	What is the role of the finally block in exception handling?	**4**	**2**	**2**
	h).	How do you import a class from a package in Java? Provide a code example.	**4**	**3**	**2**
	i).	What is the difference between String and String Buffer in Java?	**5**	**2**	**2**
	j).	Explain the purpose of the synchronized keyword in Java multithreading.	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Describe the main principles of Object-Oriented Programming.	**1**	**2**	**5**
	b).	Write a simple Java program to display "Hello, World!" and explain each line of code.	**1**	**3**	**5**
		OR			
3.	**a).**	List and explain the different data types available in Java.	**1**	**2**	**5**
	b).	What is type casting in Java? Write a program to demonstrate implicit and explicit type casting.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Explain the concept of constructors in Java. How do they differ from regular methods?	**2**	**2**	**5**
	b).	Write a Java program to create a class with overloaded constructors. Show how each constructor is called.	**2**	**3**	**5**

		OR			
5.	**a).**	Define method overriding. How is it different from method overloading?	**2**	**2**	**5**
	b).	Provide an example to demonstrate method overriding in Java.	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	Describe the process of declaring and initializing a one-dimensional array in Java.	**3**	**3**	**5**
	b).	Write a Java program to find the maximum element in an array of integers.	**3**	**3**	**5**
		OR			
7.	**a).**	Explain the concept of inheritance in Java. What are the different types of inheritance supported by Java?	**3**	**2**	**5**
	b).	Write a Java program to demonstrate multilevel inheritance.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	What are packages in Java? Why are they used?	**4**	**2**	**5**
	b).	Create a package named com.example and a class named Hello within this package. Write a program to display "Hello, Package!".	**4**	**3**	**5**
		OR			
9.	**a).**	Describe the try-catch-finally mechanism in Java exception handling.	**4**	**2**	**5**
	b).	Write a Java program that demonstrates handling multiple exceptions using multiple catch blocks.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Explain the differences between String, StringBuilder, and String Buffer.	**5**	**2**	**5**
	b).	Write a Java program to reverse a string using StringBuilder.	**5**	**3**	**5**
		OR			
11.	**a).**	What is JDBC? Describe its architecture.	**5**	**2**	**5**
	b).	Write a Java program to establish a connection to a MySQL database and execute a simple query to retrieve data from a table.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

SRKR ENGINEERING COLLEGE AUTONOMOUS Estd. 1980

					Course Code:B23IT2103
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
II B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER ORGANIZATION					
Common to IT & CSBS					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	**KL**	**M**
1.	**a).**	Convert the function to another canonical form. F(x,y,z)=π(0,3,6,7)	1	**3**	**2**
	b).	What is RTL?	1	**1**	**2**
	c).	List out computer registers	2	**1**	**2**
	d).	Explain Subtraction of Signed Numbers with example	2	**2**	**2**
	e).	Represent F=(A+B)*(C+D) in two-address instruction format	3	**3**	**2**
	f).	Explain CAR	3	**1**	**2**
	g).	Define memory read and write operation	4	**1**	**2**
	h).	What is hit ratio?	4	**1**	**2**
	i).	What is the need of Interface	5	**1**	**2**
	j).	Isolated I/0 vs memory mapped I/O	5	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Simplify F(A,B,C,D)=∑(1,2,3,6,7,10,12,13)	**1**	**3**	**5**
	b).	Construct Arithmetic and Logic shift unit	**1**	**3**	**5**
		OR			
3.	**a).**	Construct JK Flip flop along with truth table	**1**	**3**	**5**
	b).	Explain about Bus and Memory Transfer	**1**	**2**	**5**
		UNIT-2			
4.	**a).**	Design Binary Adder and Subtractor and explain	**2**	**3**	**5**
	b).	Describe Instruction cycle in computer system	**2**	**2**	**5**
		OR			
5.	**a).**	Explain about Computer instructions	**2**	**2**	**5**
	b).	Describe Signed-operand Multiplication with example	**2**	**2**	**5**
		UNIT-3			
6.	**a).**	Explain about General register organization with seven registers	**3**	**2**	**5**

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	b).	Discuss about functionality of of Micro programmed Control unit?	**3**	**2**	**5**
		OR			
7.	**a).**	Describe the Addressing Modes	**3**	**2**	**5**
	b).	Hardwired control Vs Micro programmed control	**3**	**2**	**5**
		UNIT-4			
8.	**a).**	Explain Associative Memory	**4**	**2**	**5**
	b).	Explain Memory Mapping Techniques of Cache Memory	**4**	**2**	**5**
		OR			
9.	**a).**	Illustrate Virtual Memory	**4**	**2**	**5**
	b).	How many 128 × 8 RAM chips are needed to provide a memory capacity of 2048 bytes? B. How many lines of the address bus must be used to access 2048 byte of memory? How many of these lines will be common to all chips? C. How many lines must be decoded for chip select? Specify the size of the decoders?	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Explain about Asynchronous Communication interface with neat diagram	**5**	**2**	**5**
	b).	Explain about priority interrupts and interrupts cycle	**5**	**2**	**5**
		OR			
11.	**a).**	Demonstrate Direct Memory Access	**5**	**2**	**5**
	b).	Explain daisy chain priority interrupt	**5**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

SRKR ENGINEERING COLLEGE AUTONOMOUS

Course Code: B23HS2203	
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)	**R23**
II B.Tech. II Semester MODEL QUESTION PAPER	
OPTIMIZATION TECHNIQUES	
(Common to AIML & IT)	
Time: 3 Hrs.	**Max. Marks: 70 M**
Answer Question No.1 compulsorily	
Answer **ONE Question** from **EACH UNIT**	
Assume suitable data if necessary	
10 x 2 = 20 Marks	

			CO	KL	M
1.	**a).**	State the optimization problem	1	2	2
	b).	Classify various types of optimization problems	1	2	2
	c).	Define feasible region.	2	2	2
	d).	Define artificial variable.	2	2	2
	e).	Name the methods to determine the initial basic feasible solution in TP.	3	2	2
	f).	What is degeneracy in transportation problem?	3	2	2
	g).	Define integer programming problem.	4	2	2
	h).	What are the methods used in solving integer programming problem?	4	2	2
	i).	Define critical activity.	5	2	2
	j).	What are the three main phases of project?	5	2	2
		5 x 10 = 50 Marks			
		UNIT-1			
2.		Solve the following nonlinear programming problem, using the Lagrangian multipliers: Optimize $Z=4x_1^2+2x_2^2+x_3^2-4x_1x_2$ Subject to: $x_1+x_2+x_3=15$; $2x_1-x_2+2x_3=20$ and $x_1,x_2,x_3>0$	1	3	10
		OR			
3.		Use the Kuhn-Tucker conditions to solve the following nonlinear programming problem: Maximize $Z=2x_1-x_1^2+x_2$ Subject to the constraints: $2x_1+3x_2 \leq 6$ $2x_1+x_2 \leq 4$ and $x_1, x_2 \geq 0$	1	3	10
		UNIT-2			
4.		Solve the following LPP by graphical method	2	3	10

<table>
<tr><td></td><td></td><td>Maximize $Z = 8x_1 + 6x_2$
Subject to the constraints: $4x_1 + 2x_2 \leq 60$
$2x_1 + 4x_2 \leq 48$ and $x_1, x_2 \geq 0$</td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>5.</td><td></td><td>Solve the following LPP by Big-M method
Minimize $Z = 4x_1 + 3x_2$
Subject to the constraints: $2x_1 + x_2 \geq 10$
$-3x_1 + 2x_2 \leq 6$
$x_1 + x_2 \geq 6$ and $x_1, x_2 \geq 0$</td><td>2</td><td>3</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-3</td><td></td><td></td><td></td></tr>
<tr><td>6.</td><td></td><td>Solve the following transportation problem.
<table>
<tr><td>Destination →
Origin ↓</td><td>D_1</td><td>D_2</td><td>D_3</td><td>D_4</td><td>Supply</td></tr>
<tr><td>O_1</td><td>5</td><td>3</td><td>6</td><td>2</td><td>19</td></tr>
<tr><td>O_2</td><td>4</td><td>7</td><td>9</td><td>1</td><td>37</td></tr>
<tr><td>O_3</td><td>3</td><td>4</td><td>7</td><td>5</td><td>34</td></tr>
<tr><td>Demand</td><td>16</td><td>18</td><td>31</td><td>25</td><td>90</td></tr>
</table></td><td>3</td><td>3</td><td>10</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>7.</td><td></td><td>Find the initial solution by VAM and optimality by MODI method for the following transportation problem.
<table>
<tr><td>Warehouse →
Factory ↓</td><td>W_1</td><td>W_2</td><td>W_3</td><td>W_4</td><td>Capacity (Supply)</td></tr>
<tr><td>F_1</td><td>21</td><td>16</td><td>25</td><td>13</td><td>11</td></tr>
<tr><td>F_2</td><td>17</td><td>18</td><td>14</td><td>23</td><td>13</td></tr>
<tr><td>F_3</td><td>32</td><td>27</td><td>18</td><td>41</td><td>19</td></tr>
<tr><td>Requirement (Demand)</td><td>6</td><td>10</td><td>12</td><td>15</td><td>43</td></tr>
</table></td><td>3</td><td>3</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-4</td><td></td><td></td><td></td></tr>
<tr><td>8.</td><td></td><td>Use Branch and Bound algorithm to solve the following problem.
Maximize $Z = 7x_1 + 9x_2$
Subject to the constraints: $-x_1 + 3x_2 \leq 6$
$7x_1 + x_2 \leq 35$
and $x_1, x_2 \geq 0$ and are integers.</td><td>4</td><td>3</td><td>10</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>9.</td><td></td><td>Use Gomory's cutting plane algorithm to solve the following problem.
Maximize $Z = x_1 + x_2$
Subject to the constraints: $7x_1 - 5x_2 \leq 7$</td><td>4</td><td>3</td><td>10</td></tr>
</table>

<table>
<tr><td></td><td></td><td>$-12x_1 + 15x_2 \leq 7$
and $x_1, x_2 \geq 0$ and are integers.</td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-5</td><td></td><td></td><td></td></tr>
<tr><td>10.</td><td></td><td>The following are the time estimates and the precedence relationships of the activities in a project network:
<table>
<tr><td>Activity</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td><td>K</td></tr>
<tr><td>Predecessor activity</td><td>-</td><td>-</td><td>-</td><td>A</td><td>B</td><td>B</td><td>C</td><td>E</td><td>D</td><td>F,G</td><td>H,I</td></tr>
<tr><td>Timeestimate (weeks)</td><td>4</td><td>7</td><td>3</td><td>6</td><td>4</td><td>7</td><td>6</td><td>10</td><td>3</td><td>4</td><td>2</td></tr>
</table>
Draw the project network diagram. Determine the critical path and the project completion time.</td><td>5</td><td>3</td><td>10</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>11.</td><td></td><td>The time estimates (in weeks) and other characteristics of a project are given below.
<table>
<tr><td>Activity</td><td>1-2</td><td>1-6</td><td>2-3</td><td>2-4</td><td>3-5</td><td>4-5</td><td>6-7</td><td>5-8</td><td>7-8</td></tr>
<tr><td>Optimistic time</td><td>3</td><td>2</td><td>6</td><td>4</td><td>8</td><td>3</td><td>3</td><td>2</td><td>8</td></tr>
<tr><td>Most likely time</td><td>6</td><td>5</td><td>12</td><td>5</td><td>11</td><td>7</td><td>9</td><td>4</td><td>16</td></tr>
<tr><td>Pessimistic time</td><td>9</td><td>8</td><td>18</td><td>6</td><td>14</td><td>11</td><td>15</td><td>6</td><td>18</td></tr>
</table>
i) Find expected activity times and their variances.
ii) Calculate expected time and variance of the critical path.</td><td>5</td><td>3</td><td>10</td></tr>
</table>

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23BS2201
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
II B.Tech. II SEMESTER MODEL QUESTION PAPER					
PROBABILITY AND STATISTICS					
Common For AIML, CSE, CSBS, CSIT & IT					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 x 2 = 20Marks
			CO	**KL**	**M**
1.	**a)**	Explain briefly the types of variables.	**1**	**2**	**2**
	b)	List out the measures of central tendency.	**1**	**1**	**2**
	c)	State the properties of distribution function.	**2**	**1**	**2**
	d)	If A and B be two events with $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$. Find (i) $P\left(\frac{A}{B}\right)$ and (ii) $P\left(\frac{B}{A}\right)$	**2**	**3**	**2**
	e)	Write the formula, to find the angle between two regression lines.	**3**	**1**	**2**
	f)	Write the normal equations to fit a curve $y = ae^{bx}$.	**3**	**1**	**2**
	g)	Write confidence limits for the population variance of a small sample?	**4**	**2**	**2**
	h)	State Central limit theorem.	**4**	**1**	**2**
	i)	Define Type I and Type II errors.	**5**	**1**	**2**
	j)	State the properties of χ^2- distribution.	**5**	**1**	**2**
					5*10=50 Marks
		UNIT-1			
2.	**a)**	Explain the methods of collecting primary and secondary data.	**1**	**2**	**5**
	b)	From the prices X and Y of shares A and B respectively given below, Compute which share is more stable in price. X: 55, 54, 52, 53, 56, 58, 52, 50, 51, 49 Y: 108, 107, 105, 105, 106, 107, 104, 103, 104, 101	**1**	**3**	**5**
		OR			
3.	**a)**	The first four moments of the random variable about origin are: -1.5, 17, -30 and 108. Find the moments about mean, β_1 and β_2.	**1**	**3**	**5**
	b)	Find the mean, median and mode for the following data. Mid value: 15, 20, 25, 30, 35, 40, 45, 50 Frequency: 2, 22, 19, 14, 3, 4, 6, 1	**1**	**3**	**5**

Table for 2 b):

X	55	54	52	53	56	58	52	50	51	49
Y	108	107	105	105	106	107	104	103	104	101

Table for 3 b):

Mid value	15	20	25	30	35	40	45	50
Frequency	2	22	19	14	3	4	6	1

<table>
<tr><td></td><td></td><td>UNIT-2</td><td></td><td></td><td></td></tr>
<tr><td>4.</td><td>a)</td><td>State and prove Bayes theorem.</td><td>2</td><td>3</td><td>5</td></tr>
<tr><td></td><td>b)</td><td>Let X be a random variable with the following probability distribution
<table><tr><td>X</td><td>6</td><td>9</td><td>12</td></tr><tr><td>P(X=x)</td><td>1/6</td><td>½</td><td>1/3</td></tr></table>Find E(x) and V(x) using the laws of expectations.</td><td>2</td><td>3</td><td>5</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>5.</td><td>a)</td><td>A manufacturer of blades knows that 5% of his product is defective. If he sells blades in boxes of 100 and guarantees that not more than 10 blades will be defective. Find the probability that a box will fail to meet the guaranteed quality?</td><td>2</td><td>3</td><td>5</td></tr>
<tr><td></td><td>b)</td><td>In a distribution exactly normal, 11.03% of the items are under 25-kilogram weight and 89.97% of the items are under 70-kilogram weight.What are the mean and standard deviation the distribution.</td><td>2</td><td>3</td><td>5</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-3</td><td></td><td></td><td></td></tr>
<tr><td>6.</td><td>a)</td><td>If the regression lines are $2y - x = 50$ and $3y - 2x = 10$, then find the mean values of x, y and the correlation coefficient between x and y.</td><td>3</td><td>3</td><td>5</td></tr>
<tr><td></td><td>b)</td><td>The marks obtained by 10 students in Mathematics(X) and Statistics(Y) are given below.
<table><tr><td>X</td><td>68</td><td>64</td><td>75</td><td>50</td><td>64</td><td>80</td><td>75</td><td>40</td><td>55</td><td>64</td></tr><tr><td>Y</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>70</td></tr></table>Find the rank correlation coefficient between the subjects.</td><td>3</td><td>3</td><td>5</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>7.</td><td></td><td>Obtain a second-degree polynomial from the following data.
<table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Y</td><td>12</td><td>10.5</td><td>10</td><td>8</td><td>7</td><td>8</td><td>7.5</td><td>8.5</td><td>9</td></tr></table></td><td>3</td><td>3</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-4</td><td></td><td></td><td></td></tr>
<tr><td>8.</td><td>a)</td><td>A random sample of 10 ball bearings produced by a company have a mean diameter of 0.5060 cm with s.d. 0.004 cm. Find the maximum error estimate E and 95% confidence interval for the actual mean of ball bearings produced by this company assuming sampling from normal population.</td><td>4</td><td>3</td><td>5</td></tr>
<tr><td></td><td>b)</td><td>Determine a 95% confidence interval for the variance of a normal population with the sample: 145.3, 145.1, 145.4 and 146.2.</td><td>4</td><td>3</td><td>5</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>9.</td><td></td><td>A population consists of five numbers 2, 3, 6, 8 and 11, Consider all possible samples of size two which can be drawn with replacement for the population, Calculate (a) The mean of the population (b) The standard deviation of the population (c) The mean of the sampling distribution of means (d) The standard deviation of the sampling distribution of means.</td><td>4</td><td>3</td><td>10</td></tr>
</table>

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<table>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-5</td><td></td><td></td><td></td></tr>
<tr><td>10.</td><td>a)</td><td>In a large city A, 20 percent of a random sample of 900 school children had defective eye-sight. In other large city B, 15 percent of random sample of 1600 children had the same defect. Is this difference between the two proportions significant?</td><td>5</td><td>3</td><td>5</td></tr>
<tr><td></td><td>b)</td><td>A group of 10 boys fed on a diet A and another group of 8 boys fed on a different diet B recorded the following increase in weights.
Diet A (in Kg.) : 5 6 8 1 12 4 3 9 6 10
Diet B (in Kg.): 2 3 6 8 10 1 2 8
Test whether the variance of weights differ significantly?</td><td>5</td><td>3</td><td>5</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>11.</td><td></td><td>Fit a Poisson distribution to the given data and test the goodness of fit.
<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>f</td><td>275</td><td>72</td><td>30</td><td>7</td><td>5</td><td>2</td><td>1</td></tr></table></td><td>5</td><td>3</td><td>10</td></tr>
</table>

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980 SRKR ENGINEERING COLLEGE AUTONOMOUS

Course Code: B23IT2201

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R23**

II B.Tech. II Semester MODEL QUESTION PAPER

ADVANCED DATA STRUCTURES & ALGORITHMS

For IT

Time: 3 Hrs. | **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	**KL**	**M**
1.	**a).**	Write any two applications of B-tree?	**1**	**2**	**2**
	b).	Define an articulation point?	**1**	**1**	**2**
	c).	Define Convex hull problem?	**2**	**1**	**2**
	d).	Explain the control abstraction of divide and conquer?	**2**	**2**	**2**
	e).	What is job sequencing with deadlines problem?	**3**	**1**	**2**
	f).	Define principle of optimality?	**3**	**1**	**2**
	g).	Define Sum of Subsets problem?	**3**	**1**	**2**
	h).	Explain about Least Cost Search?	**3**	**2**	**2**
	i).	Define P and NP classes	**4**	**1**	**2**
	j).	What is satisfiability Problem	**4**	**1**	**2**

5 x 10 = 50 Marks

		UNIT-1			
2.	**a).**	What are the applications of AVL-tree?	**1**	**1**	**3**
	b).	Construct a B-Tree of order 4 for the following elements 5, 3, 21, 9, 13, 22, 7, 10, 11, 14, 8, 16	**1**	**3**	**7**
		OR			
3.	**a).**	Construct a Max Heap for the following list of elements 23, 45, 65, 12, 43, 67, 73, 87, 34, 88.	**1**	**3**	**3**
	b).	Construct an AVL Tree for the following list of elements 45, 89, 95, 36, 29, 20, 28, 34.	**1**	**3**	**7**
		UNIT-2			
4.	**a).**	Apply articulation algorithm to find the articulation points in the given graph	**1**	**3**	**5**

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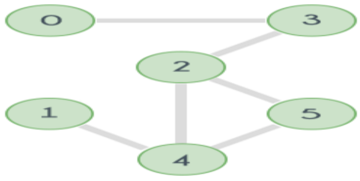
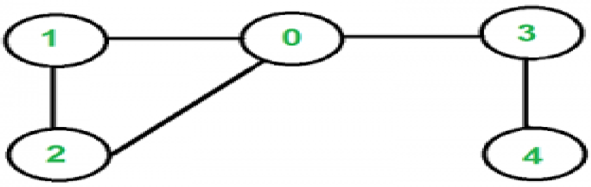
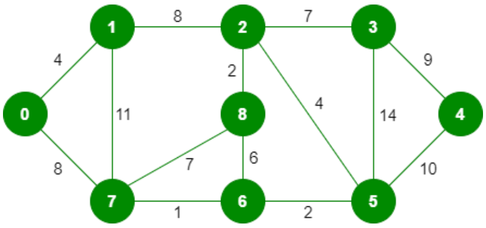
		 Fig. 2			
	b).	Apply quick sort algorithm to sort the following list of elements 5, 3, 1, 9, 8, 2, 4, 7	2	3	5
		OR			
5.	a).	Write an algorithm for BFS and apply BFS algorithm to traverse in the following graph. 	1	3	5
	b).	Apply Divide and conquer technique, explain strassen's matrix multiplication?	2	3	5
		UNIT-3			
6.	a).	Apply the greedy method, to find the optimal solution for following of the knapsack problem n = 3, m = 20 (P1, P2, P3) = (25, 24, 15) and (W1, W2, W3) = (18, 15, 10).	3	3	5
	b).	Apply kruskal's algorithm, to find the minimum cost spanning tree for the given graph 	3	3	5
		OR			
7.	a).	Construct an OBST for c(i, j) $0 \leq i \leq j \leq 4$ the identifier set (a1, a2, a3, a4) = (do, if, int, while) with (p1, p2, p3, p4) = (3, ,3, 1, 1) and q(0:4) = (2, 3, 1, 1, 1)	3	3	10
		UNIT-4			
8.	a).	Solve the following sum of subsets problem n=6, M=30 and W(1…6)=(5,10,12,13,15,18) using Backtracking	3	3	10
		OR			
9.	a).	Solve the following 0/1 knapsack problem using LCBB. n = 4, (P1, P2, P3, P4) = (10, 10, 12, 18) and (W1, W2, W3, W4) = (2, 4, 6, 9) and m = 15.	3	3	10

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		UNIT-5			
10.	**a).**	Explain the classes of NP- Hard and NP-Complete	**4**	**2**	**5**
	b).	Prove that clique Decision Problem is NP-Complete	**4**	**3**	**5**
		OR			
11.	**a).**	Explain about Job Shop Scheduling	**4**	**2**	**5**
	b).	Prove cook's theorem	**4**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980 SRKR ENGINEERING COLLEGE AUTONOMOUS

					Course Code: B23IT2202
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R23**
		II B.Tech. II Semester MODEL QUESTION PAPER			
		SOFTWARE ENGINEERING			
		Common to IT & CSBS			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					10 x 2 = 20 Marks
			CO	**KL**	**M**
1.	**a).**	Define software engineering.	**1**	**1**	**2**
	b).	List out the framework activities.	**1**	**1**	**2**
	c).	List any two risk management strategies.	**2**	**2**	**2**
	d).	State the Software Requirements Specification?	**2**	**1**	**2**
	e).	Define data flow diagram.	**3**	**2**	**2**
	f).	State types of user interfaces?	**3**	**2**	**2**
	g).	What is system testing?	**4**	**1**	**2**
	h).	What is the need of Software Quality Assurance?	**4**	**1**	**2**
	i).	What is the advantage of software maintenance.	**5**	**1**	**2**
	j).	Define reuse approach?	**5**	**1**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Define the term "software engineering" and explain its significance in the evolution of computer system engineering.	**1**	**2**	**5**
	b).	Explain waterfall model with neat diagram.	**1**	**2**	**5**
		OR			
3.	**a).**	Explain spiral model.	**1**	**2**	**5**
	b).	Write about agile development model.	**1**	**2**	**5**
		UNIT-2			
4.	**a).**	Determine the strengths and weaknesses of various project estimation techniques?	**2**	**3**	**5**
	b).	Differentiate between project estimation techniques, empirical estimation techniques.	**2**	**3**	**5**
		OR			
5.	**a).**	Define axiomatic specification and algebraic specification. Explain	**2**	**2**	**5**
	b).	Create SRS document for “Online Job Portal”.	**2**	**3**	**5**

SRKR ENGINEERING COLLEGE AUTONOMOUS Estd. 1980

		UNIT-3			
6.	**a).**	Use the characteristics that define a good software design. How does it impact maintainability, scalability, and performance?	**3**	**3**	**5**
	b).	Determine the core principles and values of agile methodology.	**3**	**3**	**5**
		OR			
7.	**a).**	Determine the SA/SD (Structured Analysis/Structured Design) methodology.	**3**	**3**	**5**
	b).	How do the key characteristics of user interface contribute to user satisfaction and usability?	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Determine the key components of good software documentation?	**4**	**3**	**5**
	b).	Interpret the key differences between black-box, white-box.	**4**	**3**	**5**
		OR			
9.	**a).**	Determine any five common challenges faced when applying statistical testing in software quality assurance?	**4**	**3**	**5**
	b).	Demonstrate the Capability Maturity Model (CMM) framework activities.	**4**	**3**	**5**
		UNIT-5			
10.	**a)**	Interpret the software reverse engineering. And list out the advantages of reverse engineering.	**5**	**3**	**5**
	b)	Interpret Several Key Aspects of Software Maintenance	**5**	**3**	**5**
		OR			
11.		Compute the methods used for estimating software maintenance costs. What factors can impact the accuracy of these estimates?	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

SRKR ENGINEERING COLLEGE

					Course Code:B23IT2203
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
II B.Tech. II Semester MODEL QUESTION PAPER					
OPERATING SYSTEMS					
Common to IT & CSBS					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	**KL**	**M**
1.	**a).**	Define Operating system.	**1**	**1**	**2**
	b).	Explain Debugging	**1**	**2**	**2**
	c).	List Multi-threading Models	**2**	**1**	**2**
	d).	Explain Context-Switching	**2**	**2**	**2**
	e).	What is Monitor?	**3**	**2**	**2**
	f).	Demonstrate Swapping	**3**	**2**	**2**
	g).	What are Contiguous memory allocation techniques.	**4**	**2**	**2**
	h).	What is Page fault?	**4**	**2**	**2**
	i).	List any two File attributes.	**5**	**1**	**2**
	j).	Define Access Matrix.	**5**	**1**	**2**
					5 x 10 = 50 Marks
		UNIT-1	**CO**	**KL**	**M**
2.	**a).**	Draw the Micro Kernal Operating System Structure and explain the components.	**1**	**2**	**5**
	b).	Describe the major services provided by Operating System	**1**	**2**	**5**
		OR			
3.	**a).**	Explain the different process management system calls	**1**	**2**	**5**
	b).	Illustrate the Booting process of Operating System	**1**	**2**	**5**
		UNIT-2			
4.	**a).**	Illustrate Thread scheduling algorithms	**2**	**2**	**4**
	b).	Assume you have the following jobs to execute with one processor, with the jobs arriving in the order listed here	**2**	**3**	**6**

Process	Burst Time	Arrival Time
P0	80	0

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<table>
<tr><td></td><td></td><td>
<table>
<tr><td>P1</td><td>20</td><td>10</td></tr>
<tr><td>P2</td><td>10</td><td>10</td></tr>
<tr><td>P3</td><td>20</td><td>80</td></tr>
<tr><td>P4</td><td>50</td><td>85</td></tr>
</table>
Find out the following by using Round Robin Scheduling Algorithm with a Quantum of 15.
i. Gantt chart illustrating the execution of these processes
ii. Turnaround time of the Processes
iii. Average wait time for the Processes
</td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>5.</td><td></td><td>
Assume you have the following jobs to execute with one processor
<table>
<tr><th>Process Id</th><th>Arrival time</th><th>Burst time</th><th>Priority</th></tr>
<tr><td>P1</td><td>8</td><td>6</td><td>1</td></tr>
<tr><td>P2</td><td>5</td><td>15</td><td>2</td></tr>
<tr><td>P3</td><td>4</td><td>8</td><td>0</td></tr>
<tr><td>P4</td><td>3</td><td>5</td><td>4</td></tr>
<tr><td>P5</td><td>0</td><td>13</td><td>3</td></tr>
</table>
Find out Average Waiting time and Average Turnaround Time using Shortest Job First (Both Preemptive and Non-Premptive)
</td><td>2</td><td>3</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-3</td><td></td><td></td><td></td></tr>
<tr><td>6.</td><td>a).</td><td>Illustrate the Critical Section Problem solution for the Dining Philosophers problem using Semaphores</td><td>3</td><td>2</td><td>5</td></tr>
<tr><td></td><td>b).</td><td>Describe in detail Sleep and Wakeup</td><td>3</td><td>2</td><td>5</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>7.</td><td></td><td>Describe Dead lock avoidance using Banker's Algorithm with an example.</td><td>3</td><td>2</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>UNIT-4</td><td></td><td></td><td></td></tr>
<tr><td>8.</td><td>a).</td><td>Compare Contiguous and Non-Contiguous memory allocations</td><td>4</td><td>2</td><td>4</td></tr>
<tr><td></td><td>b).</td><td>Find out the seek distance of read write head using Shortest seek time first Disk scheduling algorithm, when the order of cylinder request is 82, 170, 43, 140, 24, 16, 190 and current position of Read/Write head is: 50.</td><td>4</td><td>3</td><td>6</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>9.</td><td>a).</td><td>Consider the following page reference string:
1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6.
Find out the Number of Page faults using
i) FIFO, LRU and Optimal page replacement algorithms with 3 frames.</td><td>4</td><td>3</td><td>6</td></tr>
</table>

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	b).	Explain any two RAID Levels with diagram.	**4**	**2**	**4**
		UNIT-5			
10.	**a).**	Discuss File system implementation and explain File allocation algorithm.	**5**	**2**	**5**
	b).	What is the Access Matrix? Explain copy, owner and control rights with an example access matrix.	**5**	**2**	**5**
		OR			
11.	**a).**	Explain various file access methods with suitable examples.	**5**	**2**	**5**
	b).	Explain Protection goals and principles.	**5**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

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